Revisiting the Systematicity in Negation in the Era of In-Context Learning

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OVERVIEW: DO LLMS SYSTEMATICALLY UNDERSTAND NEGATION?

- LLMs often struggle with understanding negation despite massive scale. Performance is inconsistent across benchmarks.
- Lack of **systematicity** is often mentioned as a root cause.
- Previous studies focused on **behavioral outputs** (accuracy on benchmarks) rather than **internal representations**.

This study is to analyze LLMs' understanding of negation from two perspectives: behavioral systematicity and representational systematicity.

BACKGROUND: FODOR & PYLYSHYN (1988)

Systematicity Definition

"The capacity to comprehend an unbounded set of structured expressions from a finite set of primitive components."

Understanding 1

"The cat didn't chase the dog."

Implicit Ability

If you systematically understand the first sentence...

Understanding 2

You should also understand "The dog didn't chase the cat."

This implicit ability arises from linked representations of constituent concepts and their combinations.

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Classical Systems

Key Argument:

Classical systems possess structural representations that guarantee systematicity.

Connectionist Architectures

The Challenge:

Neural networks often lack these structural guarantees.
Note: The systematicity is a property of structural representations rather than just behavior.

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BEHAVIORAL VS. REPRESENTATIONAL [Vegner et al., 2025]

Behavioral Systematicity

- Observed performance patterns on test inputs.
- Evaluated via accuracy on benchmark tasks.
- Risk: Spurious correlations or memorization.

Representational Systematicity

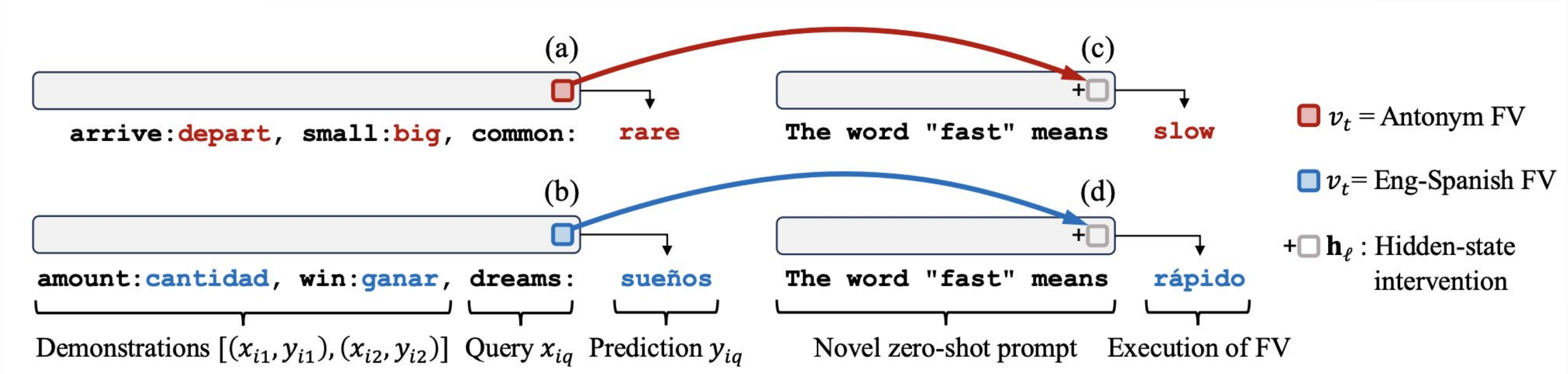
- Internal encoding of compositional structure.
- Must probe the **internal states** to verify the model's systematicity.

RESEARCH OBJECTIVE

We analyze the extent to which representations for negation understanding tasks can be systematically constructed and utilized in the in-context learning (ICL) framework of LLMs.

ANALYZING REPRESENTATIONAL SYSTEMATICITY VIA FUNCTION VECTORS

Function Vector [Todd et al., 2024]: a dense vector extracted from hidden activations that distills the **ICL mapping** demonstrated by examples.
The vector can be **injected** into zero-shot prompts to trigger the task without providing any examples.
The execution of multiple tasks (e.g. antonym prediction+capitalize) is also possible through **vector composition**.



EXTRACTING THE FUNCTION VECTOR

The vector is an aggregate activation across a set of attention heads *A* identified as having strong causal influence on the task.

$$V_{\text{func}} = \Sigma_{(l,\, j)\, \in\, A}\, \bar{a}_{l,\, j}$$

l: Layer index, j: Head index

$\bar{a}_{l,\, j}$: Mean activation over multiple prompts demonstrating the same task

Empirically, the function vector emerges in middle layers.

CAUSAL INTERVENTION BY FUNCTION VECTORS

To execute the task in a zero-shot setting, we inject V_{func} into the hidden state h_l at a selected layer.

$$h'_l = h_l + V_{func}$$

The modified state h'_l propagates through the remaining layers, inducing function execution for the task.

DEFINING THE NEGATION TASK

1. Negation Cue

Recognition of the negation expression within the sentence.
(e.g., "not", "no", "n't").

2. Negation Scope

Comprehension of the parts of the sentence affected by the negation cue.

Example input: I do not call customer service often
Negation cue: <u>not</u>
Scope: <u>call customer service often</u>

ROBUSTNESS ACROSS OUTPUT FORMATS FOR SCOPE UNDERSTANDING

We analyze four formats for the negation scope resolution task:

I do [NEG]not[/NEG] call customer service often.	
Extract	not call customer service often
Bracket	I do not [call customer service often].
Last-word	often
NER	[0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0]

EXPERIMENTAL SETUP: DATASET

SFU Review Corpus

Real-world documents across various genres.

Manually annotated at token level for cues and scope.

Filtered for sentences with one negation cue.

Sizes: 153 Train / 308 Dev / 1078 Tes
Each set is composed of different word combinations.

MODELS & EVALUATION SETTING

Selected Models

GPT-j-6B: Classic open-source LLM.

Qwen3-4B: Modern, high-performance LLM.

Baseline Task

Antonym prediction (e.g., old -> young)

Evaluation Settings

10-shot ICL: Provided for behavioral analysis.

Zero-shot + FV: Intervention with function vectors.

Zero-shot + TV: Intervention with task vectors.

Baseline: Accuracy using simple heuristics such as frequency.

BEHAVIORAL SYSTEMATICITY: OVERALL RESULTS

The accuracy is not perfect: models fail to recognize arbitrary negated sentences from demonstrations.

General performance trend: Qwen3-4B > GPT-j-6B

Task	GPT-j-6B	Qwen3-4B	Base.
Antonym (Ref)	64.5	65.1	-
Negation cue	62.0	64.3	64.7
Scope: Extract	**56.9**	**76.0**	55.7
Scope: Bracket	22.6	**69.8**	55.7
Scope: Last-word	24.5	32.6	56.9
Scope: NER	0.0	0.0	55.7

DISCUSSION: BEHAVIORAL OBSERVATIONS FOR SCOPE UNDERSTANDING

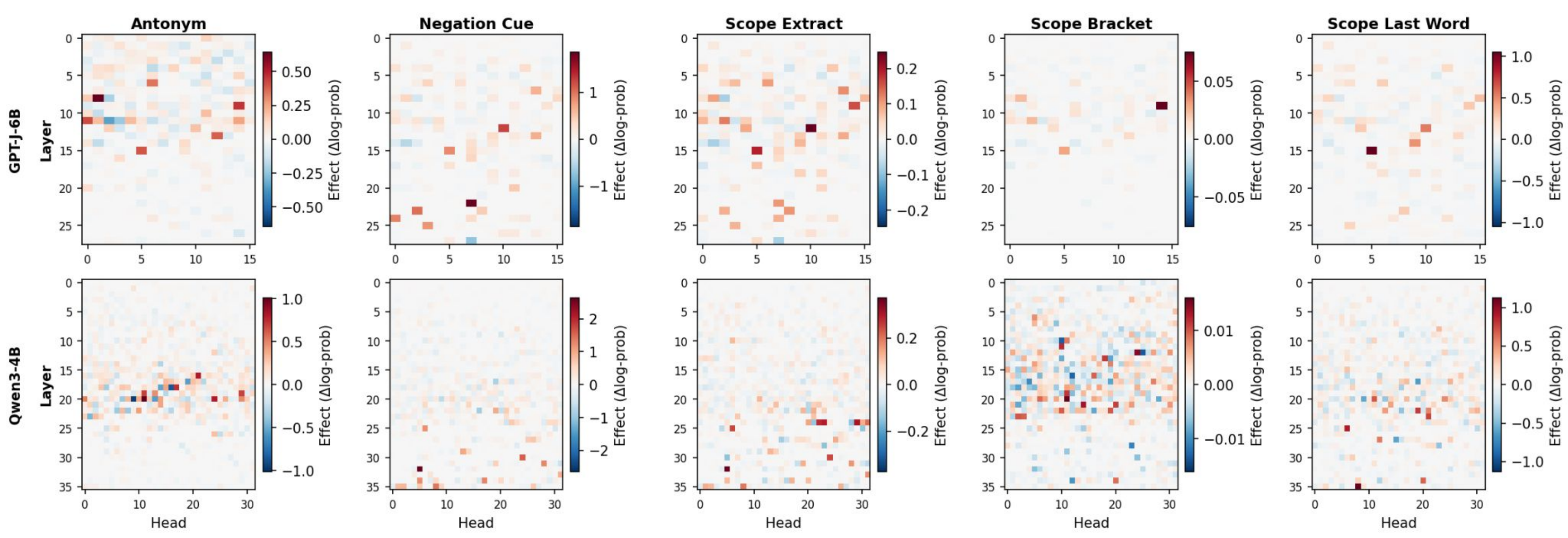
Models perform relatively well on **Extract** format (Qwen3-4B: 76.0%).

Bracket and **Last-word** formats show significant drops, especially for GPT-j-6B.

The **NER** format failed completely (0% accuracy), indicating a failure to map linguistic scope to binary structures via demonstrations.

REPRESENTATIONAL SYSTEMATICITY (FUNCTION VECTORS)

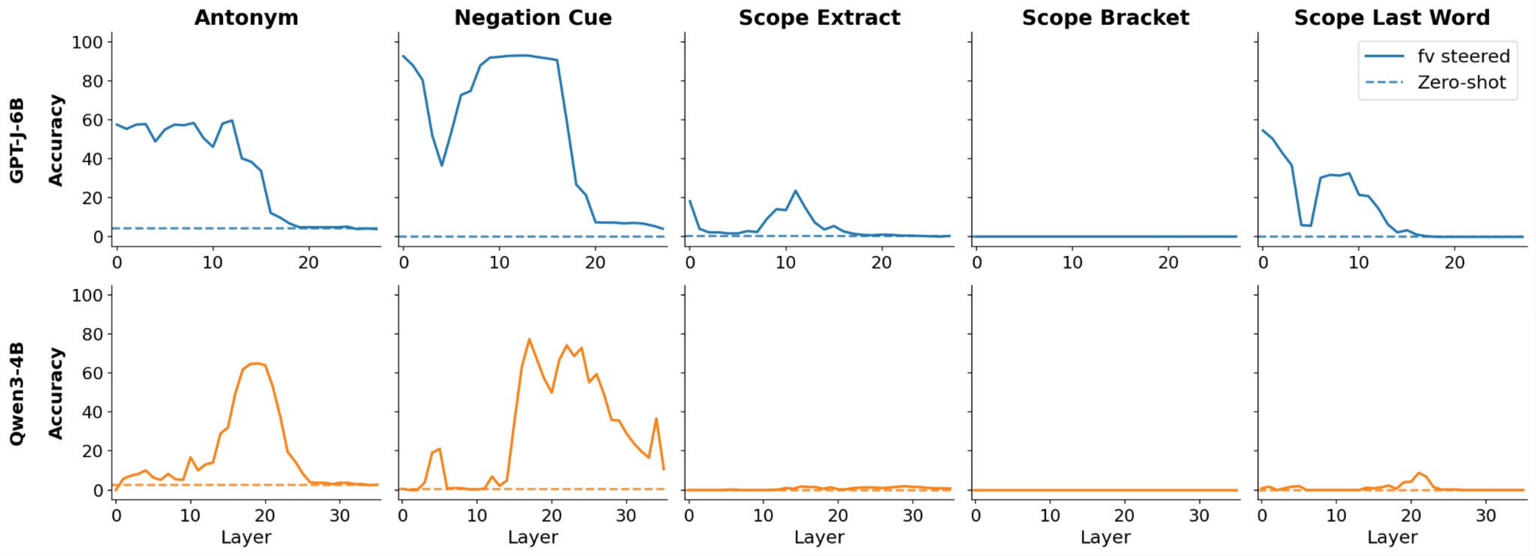
The average indirect effect across tasks for each attention head



The attention heads with high effect appear from the middle-late layers

REPRESENTATIONAL SYSTEMATICITY (FUNCTION VECTORS)

Zero-shot top-1 accuracy of applying function vectors to models



Function vectors are robust for cues, but struggle for scope

THE BOTTLENECK: SCOPE RESOLUTION

Constructing function/task vectors for negation resolution is significantly more challenging. Potential reasons:

Structure Recognition

Extracting and generating multiple tokens is harder than single token tasks.

Abstract Concepts

The "concept" of negation scope may not be captured as a linear vector transformation.

Layer Dynamics

Scope recognition might require complex, non-linear processing across layers.

MODEL COMPARISON: GPT-J VS QWEN3

GPT-j-6B

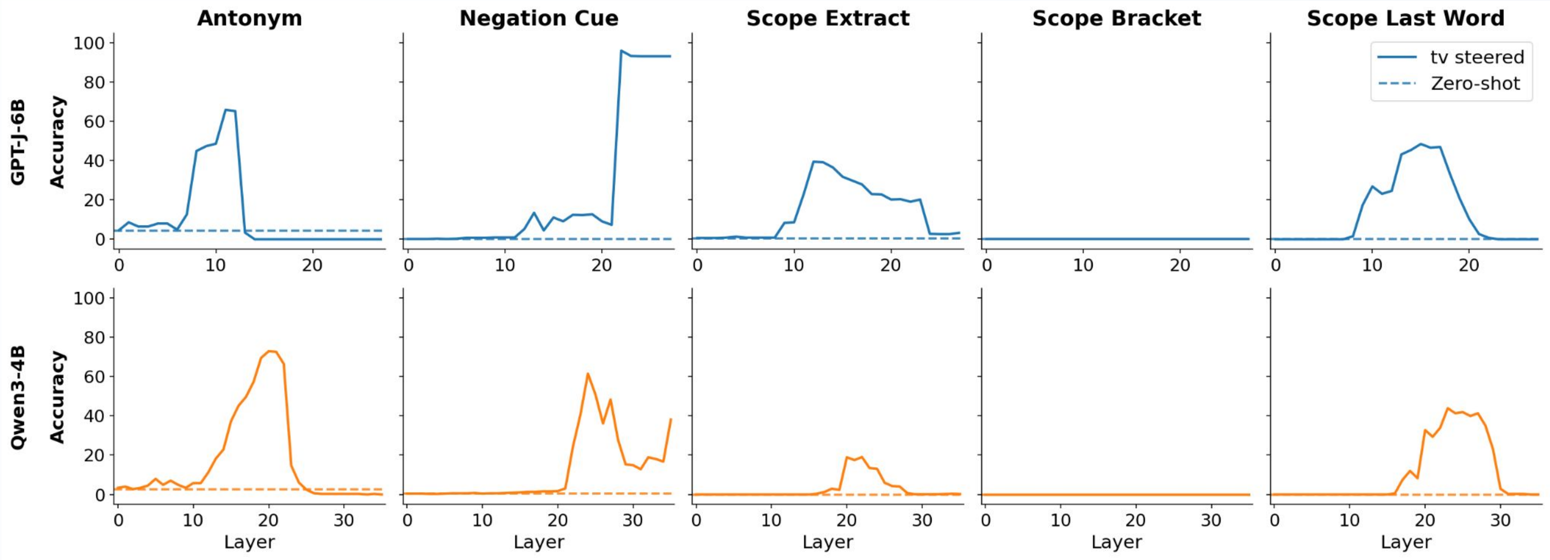
- Peaks in early-to-middle layers.
- Accuracy is low for scope.
- Representations might be shallow.

Qwen3-4B

- Peaks in middle-to-late layers.
- Higher accuracy on 10-shot settings, but the intervention effect is limited for scope.
- Representations might be more complex, but lacks systematicity for scope.

REPRESENTATIONAL SYSTEMATICITY (TASK VECTORS)

Zero-shot top-1 accuracy of applying task vectors[Hendel+2023] to models



Task vectors are more robust for cues than for scope

COMPARISON: FUNCTION VECTORS VS TASK VECTORS

Function vectors

- Early-to-middle layers for scope.
- Represent input-output functions.
- Adding function vectors cannot correct information that has already been computed in previous layers.

Task vectors

- More effective in middle-to-late layers for scope.
- Contain more global and abstract task information.
- Can overwrite past information, including earlier computational results.

CONCLUSION

Behavioral Findings: LLMs recognize negation cues and scope to some extent, but performance depends heavily on output format.

Representational Findings: While negation cue extraction is systematically represented via function vectors, scope resolution is not robustly captured.

Main Takeaway: True representational systematicity for structural linguistic concepts remains a challenge.

LIMITATIONS & FUTURE DIRECTIONS

Current Limitations

- Potential data contamination
- Scope analysis limited to post-cue occurrences
  e.g., "To Luke ten years in age difference is nothing". Restricted to single negation sentences.
- Limited function/task vector construction methods

Future Directions

- Multiple methods for constructing task-related hidden-state interventions
- Analyze complex scope and diverse negation expressions
- Extend to natural logic (monotonicity inference)